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A Multi-Track Elevator System for E-Commerce Fulfillment Centers

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***Abstract*— Fulfillment centers located in densely populated urban areas are an ever-growing need for leading online consumer websites. These urban fulfillment centers have limited land mass and must have innovative solutions to transport goods within the available vertical space. This work presents a Multi-Track Elevator (MTE) System, a competitive solution for rapid access and retrieval of goods in high-rise e-commerce fulfillment centers and warehouses. The MTE System consists of multiple vertical rails connected with angular traverse rails that allow multiple carriages to go up and down without collision. A novel turning point system switches track routes so that several carriages can move across the multiple rails for rapidly accessing many floors and collecting diverse goods. Unlike existing vertical-horizontal grid elevators and rail systems, the Roller-coaster type, self-powered carriages on the MTE system do not have to stop at switching points, but can continually move across the network of rails. This work walks through the architecture of the rail network system and techniques for switching multiple rails, followed by the design of vertical turntables for smooth, continuous rail switching. Finally, outlining the use of a simple route optimization algorithm, diverse elevator systems are compared with respect to total traveling time and distance. A proof-of-concept prototype has been built and is presented.**

Keywords: Shuttle Based Storage and Retrieval System, Logistics, Warehouse Automation, Rail System, Turntable, Multi-Robot Planning, Fulfillment Center

I. INTRODUCTION

Rapid growth of the e-commerce industry has required competition between industry leaders to provide the quickest response from the moment the customer makes an online purchase to the moment the purchase arrives at their doorstep. Currently, many e-commerce companies require large warehouses to store, package and distribute their products. These companies are facing a major technological challenge when attempting to make speedy deliveries to the center of densely populated metropolitan areas [5]. One way e-commerce companies can reduce delivery times to these areas is by locating distribution warehouses as close to the city center as possible. However, due to space limitations, in order to maintain high levels of productivity online retailers will have to take advantage of technology that can change the layout of fulfillment centers from large floor foot prints to ones with smaller footprints and increased vertical space [5].

Since the 1950s, vertical stacking and handling of materials and products has been automated with Automatic Storage and Retrieval Systems (AS/RS) consisting of aisle captive storage cranes, handling unit-loads or mini-loads (typically, pallets or bins) on shelving units in warehouses

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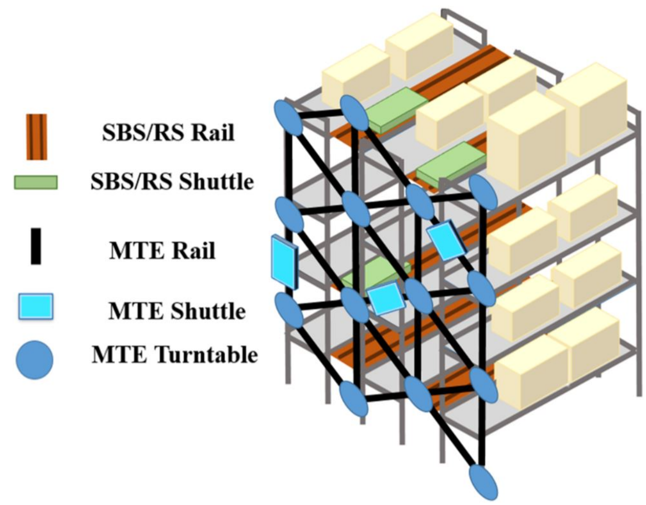

Fig. 1. An example of the implementation of the MTE system with existing warehouse automation technology

[9]. To meet the demands of increasingly small order sizes with large product variety, the Autonomous Vehicle Storage and Retrieval System, AVS/RS, was developed. The AVS/RS consists of vehicles moving horizontally along rails within the storage racks and using lifts mounted along the rack periphery to provide the vertical movement [8]. Although the AVS/RS system provides flexibility in terms of vehicle and lift allocation, it requires longer flow paths from sequential vertical and horizontal travel as well as waiting times for vehicle use of lifts [10].

Shuttle-based storage and retrieval systems (SBS/RS) were developed in an attempt to improve upon the operating capacity of the AVS/RS. In this system, two lifts capable of vertical movement of loads share a single mast to transport loads from horizontally operating shuttles to the I/O point and vice versa. Hence, the two lifts cannot pass each other, and as a result, the upper lift can only reach the Input/Output (I/O) point if the lower lift is positioned at one of the aisles below the I/O point [4].

Another advanced product sorting and handling technology used by multiple companies is the Kiva Mobile Fullfullment System (MFS), now known as Amazon Robotics. The Kiva MFS is a network of robots that are controlled by software agents that interact with each robot on a warehouse management server and on computers at human operated product picking and packing stations [12]. Amazon Robotics may be useful, but the weight and height restrictions of the shelves make this technology only a partial solution to efficiency limitations for e-commerce in cities [11].

The elevator industry has been exploring advanced architectures that could potentially be adapted for use in warehouses. One architecture of interest is a type of circular motion elevator system that consists of multiple carriages operating in a loop, similar to a metro system. This system is able to potentially increase the shaft transport capacity by up to 50 percent and can increase a buildings usable area by up to 25 percent [1]. Although the Circular Motion Elevator (CME) system provides greater efficiency than a standard elevator, the circular motion of the carriages limits where each carriage can travel and requires carriages to follow a constrained timeline. Thus, limiting the independence of each carriage to execute tasks.

Here we present a Multi-Track Elevator (MTE) system working in conjunction with an existing SBS/RS solution, shown in Fig. 1, consisting of a network of multiple tracks and self-powered carriages that can move in the vertical direction and have flexibility in horizontal movement. The carriages are designed to move along a gridded rail system, and change directions at rail intersections using vertical turntables. This system is designed to be a significantly more efficient alternative to existing systems used in warehouses.

II. DESIGN CONCEPT

A. Exploration of Rail Network Architectures

The two-dimensional grid structure, seen in Fig. 2, provides many potential path for multiple points, or in our case carriages, to travel between two nodes. Moving within a two-dimensional vertical wall, the square grid, seen in Fig. 2(a) allows each carriage to move in either vertical or horizontal directions. If oblique rails are added to the square grid, as depicted in Fig. 2(b), the net pass length moving from a grid node A to B will be shorter. In this grid structure four rails intersect at a grid node, and eight branches of rails exist at each node, giving eight choices of direction to proceed. This Octa-Grid structure creates more path opportunities, hence shorter passes moving to and from any given point on the wall. Fig. 2(c) depicts a grid architecture, where three rails intersect at each node. Horizontal rails have been eliminated in this design, which limits the directions of carriage movements from any one point to six choices. In this design, all the nodes are intersections of three rails except for the nodes at the wall boundaries.

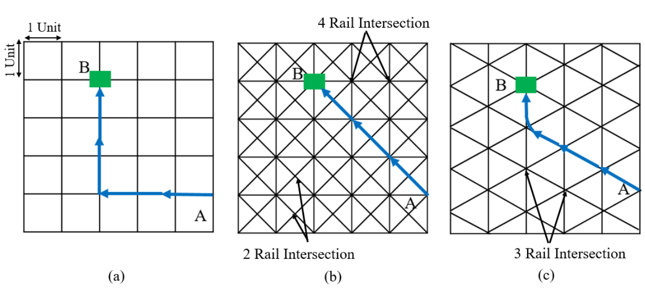

Fig. 2. Grid architecture: (a) Square-Grid (b) Octa-Grid (c) Hexa-Grid

By assuming that each block in the Square-Grid system in Fig. 2(a) is exactly one unit, the total distance traveled between arbitrary points A and B can be demonstrated. Using basic trigonometry, we can see that the total distance traveled for a carriage on the Square-Grid, Octa-Grid and Hexa-Grid architectures are 6 units, 4.24 units, and 4.6 units respectively. Although the traveling distances vary depending on the start and end points, the Square-Grid requires the longest distance, followed by the Hex-Grid and the Octa-Grid architecture; $D_{Square} > D_{Hexa} > D_{Octa}$.

It can be concluded that, although the Octa-Grid structure provides the shortest distance traveled between any two points on the grid architecture, its complexity makes it difficult to fabricate. The Vertical Hexa-Grid architecture depicted in Fig. 2(c) provides the best compromise between length of distance traveled and ease of system manufacturing.

For applications where carriages will have to visit stations that are distributed horizontally, a horizontal Hexa-Grid architecture will be more effective than the Vertical Hexa-Grid architecture, due to the avoidance of unnecessary direction changes to traverse in the horizontal direction.

B. Rail Transition

One of the most critical technical challenges in the development of two-dimensional rail systems is the realization of smooth transitions across multiple tracks and the steering of carriages at nodes where multiple tracks intersect. An existing solution is to switch two sets of grippers, one holding a horizontal rail and the other holds a vertical rail. A carriage approaching a node with one set of grippers holding a horizontal rail grasps a vertical rail at the intersection with another set of grippers, and then disengages the set of horizontal grippers. Such an approach requires a car to stop at an intersection for switching grippers, thereby taking a long time to cross over each intersection [1].

The inefficient discontinuous transition can be eliminated by applying a point switching technique. Similar to a railway turnout, a carriage can be guided from one branch of rails to another at an intersection. The wheel of a carriage is always engaged with some portion of the rail continually, and no discrete operation for engaging and disengaging grippers is involved. Unlike the traditional railway turnout that uses a pair of rail blades lying horizontally on the ground, this system is laid vertically. A turn table with multiple transition rails stands vertically and switches vertical rails.

Figure 3 presents the design concept for a Vertical Turnout Mechanism placed at the intersections of a grid rail system. The Mechanism consists of a Turntable, a set of Transition Rails, and an actuator (not shown in the Figure) that rotates the turntable. When a carriage approaches an intersection, the turntable rotates to connect the rails required for the incoming carriage's path. Figure 3 shows three transition rails secured to the turntable; a straight rail in the center, a left-curved rail, and a right-curved rail.

In case of moving straight, the turntable simply rotates to a specified angle, with the transition rail adopting the straight center configuration as seen in Fig. 3(A). When turning right or left, the positioning control of the turntable rotates to align one of the two curved rails to connect the rail of the incoming car to the desired outgoing rail as seen in Fig. 3(B) and (C), respectively. The carriage is continuously engaged with the connected rails, thus maintaining the traverse velocity. Although the three transition rails depicted in Fig. 3 allow only three available path directions, an arbitrary path across the entire two-dimensional grid is attainable by attaching the transition rails to a turntable with the ability to rotate 360°.

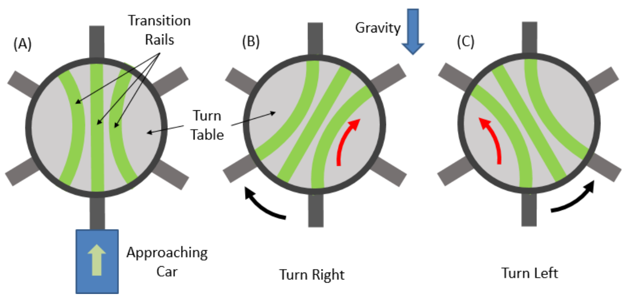

Fig. 3. The design concept for a Vertical Turnout Mechanism

III. CONTROL

A. Low-Level Control

The central controller must be able to determine the location, direction and intended path of each carriage to appropriately control the carriages and turntables. In order to execute a turn, the central controller must know the exact moment the carriage begins and ends a turn, providing appropriate carriage motor commands that adapt to the differing amounts of rack teeth on the inner and outer curved rails of the turntable. The central controller must also be aware of the impending approach of a carriage to a turntable so that the appropriate turntable position can be assumed.

The locations required to facilitate successful operation of the carriages and turntables are the two points immediately on the intersection of the turntables and rails and the point located directly at the midpoint between the two turntables. The former points will allow for successful turning commands of the carriages and the latter will provide indication of a carriages approach to a turntable with sufficient advance notice to orient the turntable to the make the appropriate rail connections.

To facilitate localization of each carriage as well as turntable orientation commands, each point used will need to be entirely unique. If the setup described previously is expanded to a larger system, such as the one depicted in Fig. 1, a substantial number of unique points will be required. Methods such as using an Internal Position System (IPS) and RFID tags for autonomous mobile robot localization are currently implemented in many warehouses and are currently undergoing extensive research [13].

An affordable alternative to RFID tags that still allows for unique identification is using infrared (IR) LEDs configured to each output a unique signal. Using a single Mojo V3 FPGA Development Board, up to 84 IR LEDs can be controlled, more than enough for the prototype depicted in Fig. 8. The IR LEDs are mounted to the inner section of the rails such that they are flush with the top of the rails. To successfully read each IR LED, each carriage is equipped with IR Receiver. As a carriage passes and reads a unique IR LED it transmits its location to the high level control program which then determines the appropriate turntable commands.

B. High Level Control

Although the focus of this paper is on the design of the system, it was necessary for testing and validation to implement a basic form of distributed path planning of the carriages. The field of robot path planning has been extensively studied [7] and divided into two main categories, centralized planning and decoupled planning. Where centralized planning considers the various robots as separate components of a composite robot and decoupled planning creates a path for each robot independently of other robots and then considers interactions among the paths [7].

Due to the large dimension of the configuration space for complicated centralized planning problems, the time complexity is exponential in this direction, a decoupled path planning approach was chosen for implementing the MTE system. One approach to decoupled planning is known as path coordination. This approach generates a free path for each robot independently and then coordinates the paths to avoid collisions. To ensure robustness of a final solution, an additional algorithm based on a priority planning approach may be executed. Additional simplifications to the requirements of the path planning problem can be made when taking into account the path constraints due to the structure of the MTE system. These physical constraints allow the system to be represented as a graph with nodes with directionality constraints.

Although there are many well known graph searching shortest-path algorithms including Dijkstras algorithm [6], A* algorithm, and the Floyd-Warshall algorithm, the physical constraints placed on the system offer some serious challenges for successful implementation using a combination of path coordination and priority planning.

For example, say we have two carriages that require shortest paths to their respective targets, although all three algorithms will easily be able to calculate the shortest path of the first carriage, they may encounter some difficulties calculating the shortest path of the second carriage that does not intersect the first carriage. It is not in the nature of these three algorithms to accept the a path that is not deemed the shortest to be the final solution. If say, carriage 2 had to extend its path to avoid carriage 1, then the three algorithms may run into a dead end if they are prohibited from returning paths that avoid collisions.

To avoid this problem, it was determined that the simplest method to implement that would avoid collisions as well as provide the overall shortest allowed path for the carriages involved was to create a modified Depth-First-Search (DFS) algorithm [3] that utilizes known decoupled path planning methods. A standard DFS allows the search to travel as deep as possible from vertex to vertex before backtracking. This modified DFS algorithm explores all independent simple possible paths that multiple carriages can travel to reach target locations, and then chooses a final path based on collision avoidance in order of priority. The modified DFS algorithm determines an initial priority queue of the carriages and begins comparing the possible shortest paths of each carriage and eliminating conflicts based on priority. At any point, if the local minimum of the shortest possible path calculated for a particular priority queue matches the overall global minimum path length, the algorithm returns the shortest path for each of the carriages, as well as the turntables and LEDs projected to be encountered by each carriage as it traverses the system.

The high level control program may be expanded so that it saves the potential paths of each carriage and can dynamically update each carriage's assigned path in the event of a change in the system. This may occur if, for example, an additional carriage is added to the MTE network whose path takes priority over another carriage, or a carriage begins to fail to the point where it may cause a collision.

IV. DESIGN IMPLEMENTATION

A. Turntable and Rail Design

To reduce manufacturing complexity, the rails of the MTE system, seen in Fig.4, between the turntables are cylindrical with a diameter 0.625 cm (0.25 in) of and modeled after standard roller coaster rails.

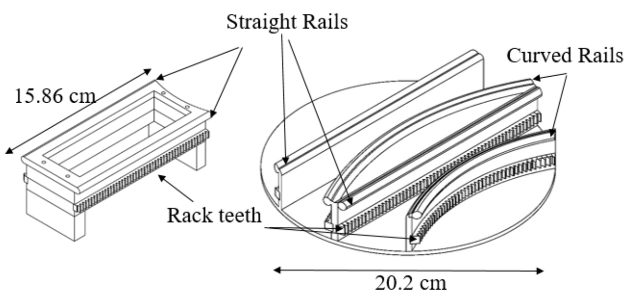

Fig. 4. The design of the rail system and transition turntable platform

A basic torque calculation was completed to determine the minimum torque required to rotate the turntable. To accommodate incoming carriages, the turntable should be able to perform a full rotation in under 2 seconds. Based on motor availability, an average speed of 30 RPM was selected for the motor calculation seen in (1).

$$\tau_{\text{motor}} = I\alpha = \frac{1}{2} m R_{\text{turntable}}^2 \alpha \tag{1}$$

Integrated motors with built-in drive and controller (Dynamixel MX-106T and MX-64) were selected due to the ease of use, reduced cable requirement, and large torque outputs (8.4 Nm and 6 Nm respectively).

Each feasible rail position on the turntable requires a rotation of 30 degrees. Common gear specifications found online were considered and compared to find a reasonable compromise between pitch and resolution. To have as precise of a resolution as possible, it was necessary to find the pitch that would provide the most gear teeth in the smallest amount of space, thus gears with a pitch of 32 and 62 teeth were selected. Although these gear teeth allow for an angular resolution of 5.8 degrees, the resolution of the Dynamixel motor is .088 degrees which results in the precise positioning of the turntable.

To eliminate backlash and ensure rail alignment at each of the required positions, a spring-loaded kinematic coupling was designed and can be seen in Fig. 6. When designing the kinematic coupling, it was imperative that the coupling be able to maintain its position when a carriage was traveling across the turntable as well as be able to release its position when force was applied by the turntable motor. The equations for calculating the kinematic coupling spring and pre-load requirements using the free body diagram seen in Fig. 5 are seen below.

$$\begin{gathered} F_{\text{preload}} = k\left(\Delta x_{\text{preload}}\right) \geq \frac{F_{\text{carriage,g}}}{3} \\ k \geq \frac{1.96N}{3\,(.0009m)} \geq 714.4\frac{N}{m} \end{gathered} \tag{2}$$

$$\begin{gathered} F_p = k\left(\Delta\left(x_{\text{preload}} + x\right)\right) \leq \frac{F_{\text{motor}}}{3} \\ k \leq \frac{97.98N}{3\,(.0009m + .0025m)} k \leq 3166\frac{N}{m} \end{gathered} \tag{3}$$

Based on spring availability, a spring that was .375 in long and had a spring constant of 1325.7 N/m was implemented into the kinematic coupling design.

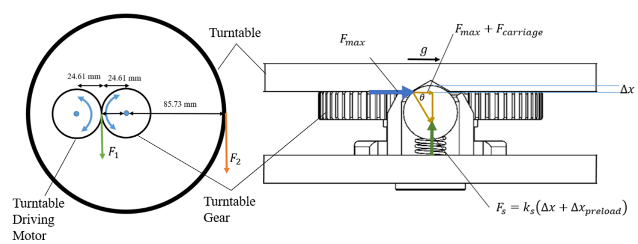

Fig. 5. The free body diagram of the turntable kinematic coupling

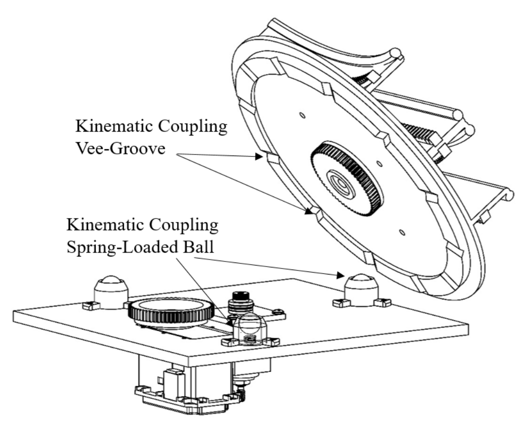

Fig. 6. The design of the rail system and transition turntable platform

To avoid Abbé errors when mounting the turntable on the wall, the wall mount, depicted in Fig. 7, was designed to distribute the moment of the turntable as it is acted upon by the carriages and gravity. A shoulder bolt is fed through the top of the turntable and kinematic coupling and rests on a thrust washer to ensure smooth turning. A set of radial bearings and a shaft bearing provide the necessary stabilizing force required to hold the turntable base plate parallel to the wall while bearing the weight of the carriages and cargo.

In the full assembly shown in Fig. 8 as well as in the supplemental video, thirteen turntables have been constructed to implement a Vertical Hexa-Grid rail structure.

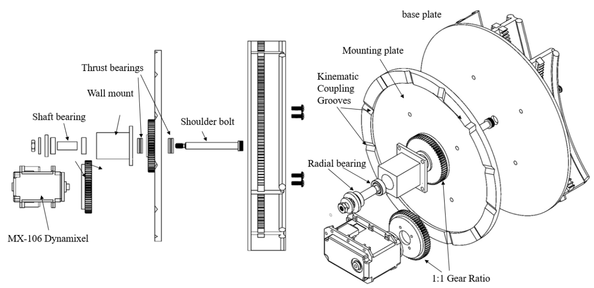

Fig. 7. An exploded view of the turntable assembly

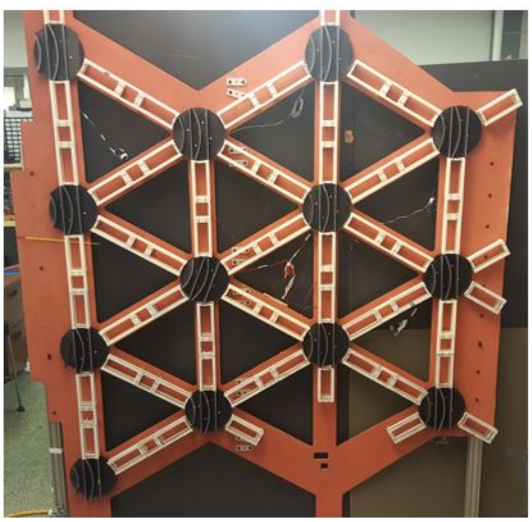

Fig. 8. The constructed realization of the turntable and rail assembly used for testing

B. Carriage Design

As seen in Fig. 9, the base of the carriage is modeled to resemble that of a roller coaster. Three sets of radial bearings act as wheels that grip the rails on three sides to constrain the carriage. To counteract the gravity load and prevent jamming, the carriage is driven by a pair of individually driven gear motors attached to a set of pinions which mate with the rack teeth on the rail architecture. In the event of a loss of power, the gear motors will prevent system from being back-drivable and locks the carriage into place.

The space constraints between the curved and straight rails on the turntable limits the size of the pinion that can be attached to the carriage. For this reason, the pinions were designed to have an pitch diameter of 0.5 in, and a DP of 24. As seen in Fig. 4, the turntable rails and connecting rails are fitted with rack teeth that are 0.625 cm thick and have a diametral pitch (DP) of 24 that are designed to mate with pinions on the carriages shown in Fig. 9.

Each carriage is predicted to weigh no more than 0.2 kg. The motors attached to each carriage were required to the specifications as calculated in (4) and (5).

$$F_g = (.2kg)\left(9.8\frac{m}{s^2}\right) = 1.96N \tag{4}$$

$$\tau_{\text{motor}} = F_g r_{\text{pinion}} = (1.96N)\,(.003175m) = .0062Nm \tag{5}$$

As seen in (5) above, very little torque is required to move such a small mass against the force of gravity. However, by adding flexibility in the weight specification of the carriage and the desire to ensure the ability to overcome small rail misalignments, the carriage motors were chosen to have a 291:1 gear ratio and a maximum output torque of 0.5 Nm with a recommended continuous load of 0.17 Nm. Even with a continuous load, the chosen carriage motors provide nearly thirty times the required torque to drive the carriage, thus allowing for a significant decrease in applied torque to increase the speed of the carriage with a continuous factor of safety.

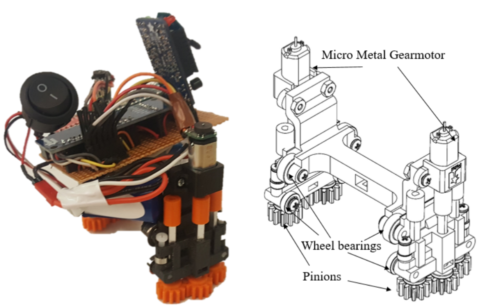

Fig. 9. A photograph of the prototype carriage design

The prototype, shown in Fig. 9 and in the supplemental video, an Arduino Micro and dual-motor driver are responsible for controlling the gearmotors. A quadrature encoder is attached to each of the motors and is used in conjunction with a PI-controller to ensure the constant velocity of each of the gearmotors.Each carriage is also equipped with a Series 1 Xbee for communication with a central controller.

V. NUMERICAL AND EXPERIMENTAL VALIDATION

A. Comparison with Current Available Elevator Systems

The design of the Vertical Hexa-Grid structure and turntable assembly was evaluated by comparing the effectiveness of this rail system to that of a Dual Standard Elevator Shaft (DSES) where two carriages share a single elevator shaft and the CME System [1]. In this context, the effectiveness of a system is defined as the normalized time required to transport cargo between two arbitrary points. The designs were also evaluated based upon the ability for carriages to transition through direction changes without stopping.

As shown in Fig. 10, one simple method for comparing the effectiveness of the different elevator systems is by assigning three arbitrary cargo transportation tasks to each of the systems. Assuming the travel speed and product handling time of the elevator carriages is virtually identical for the three systems, the efficiency depends on the architecture of rail network alone.

The modified DFS algorithm was applied to the MTE System and compared with the optimal route for the DSES and CME Systems, as shown in Figs. 10 and 11. For the three retrieval tasks the scheduling of the DSES, seen in Fig. 10(a), requires the coordination of the upper and lower elevators to avoid collisions from occurring. The timeline for the optimal efficiency of system for this operation can be seen in Fig. 11. The lower elevator reaches its destination first and begins loading the cargo as the upper elevator reaches the sixth floor. As the upper elevator is loading the cargo, the lower elevator begins its descent. As the lower elevator finishes unloading its cargo at the I/O point, and descends to the lowest level while the upper elevator approaches the I/O point. The upper elevator unloads its cargo, ascends to the the third floor to retrieve the cargo and returns to the I/O point.

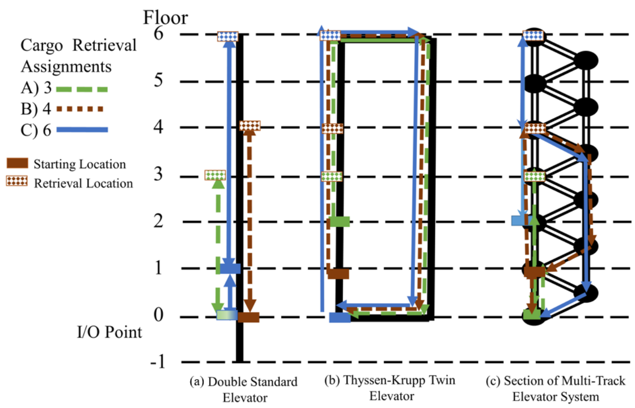

Fig. 10. Comparison of carriage travel for (a) dual standard elevator shaft, (b) the Circular Motion Elevator system and (c) the Multi-Track Elevator System

The timeline for the CME system, seen in Fig. 10(b), is shown in Fig. 11. All three elevators begin to ascend and stop at their respective floors and retrieve their cargo. The middle and lower elevators are required to wait for a short period of time as the upper elevator continues the cyclical motion and begins its descent. As each elevator descends it continues horizontally to the I/O point and unloads its cargo.

Using the MTE system, shown in Fig. 10(c), a carriage is able to pick up the cargo on the left side of the sixth floor, descend at a downward angle to the turntable on the right side of the fourth floor. When executing the second task, a carriage is able to pick up the cargo at the right turntable

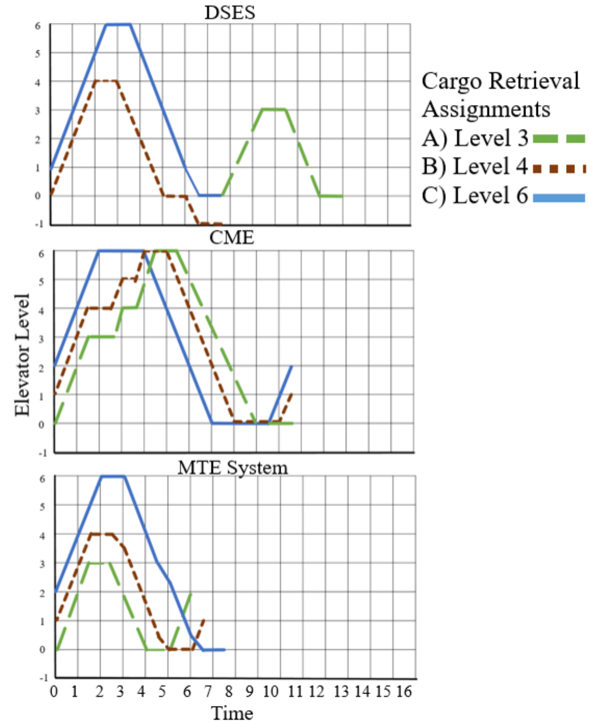

Fig. 11. Comparison of timeline for carriage travel for a dual standard elevator shaft, the CME System and the MTE System

on the fourth floor and descend to the turntable on the left side of the second floor. Finally, to complete the third task a carriage is able to pick up the cargo at the rightmost turntable on the sixth floor and descend to the rightmost turntable on the second floor without requiring a direction change. In a full-scale realization of the Multi-Track Elevator System, the Hexa-grid rail architecture will be much more expansive and will contain many more nodes, thus being able to complete the cargo transportation from a much wider array of cargo assignments.

As mentioned previously, our study assumes that the elevator cabins of all three systems travel at approximately the same speeds. Therefore, the effectiveness of each system is evaluated solely on architecture, without requiring a specific time scale. As seen in the cargo transportation comparison timeline depicted in Fig. 11 above, the effectiveness of the dual standard elevator shafts is limited by the requirement that the elevators can not cross paths. This is particularly apparent when viewing the execution of the second and third arbitrary transportation tasks. In these cases, although the lower elevator finishes its task first, it can not pass the upper elevator. The CME Elevator is more effective than the standard elevator system because it is able to transport all three cargo loads without any path interference. However, depending on the assigned task, this system still requires each elevator cabin to initially travel in a direction opposite to that of its final destination. In these instances, the multi-directionality and flexibility of the Multi-Track Elevator system allows the carriages to travel directly towards their respective final destinations while simultaneously scheduling to account for path overlapping and interference.

B. Collision Avoidance and Complex Maneuvers

The advantages of the Hexa-Grid structure and its application in logistics centers can be highlighted by observing the complex patterns and scheduling that can be executed. A common scenario, shown in Fig. 12, occurs when multiple carriages are carrying loads towards destinations that require their shortest distance paths to overlap with one another. The timeline of the complex maneuver shown in Fig. 12, indicates that although not all individual carriages take the optimal shortest path, they are able to maintain continuous motion throughout the duration of their travel while simultaneously avoiding routing collisions.

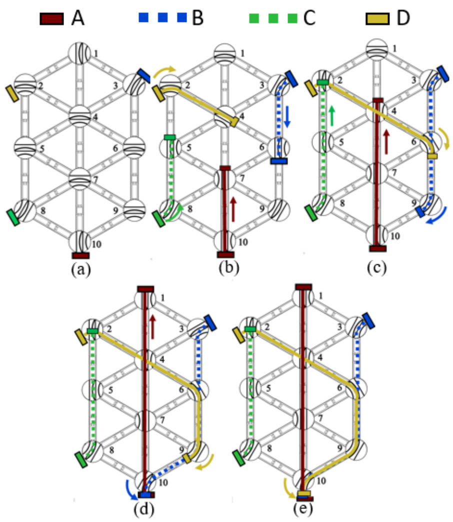

Fig. 12. A complex pattern executed by four carriages on the Multi-Track Elevator System.

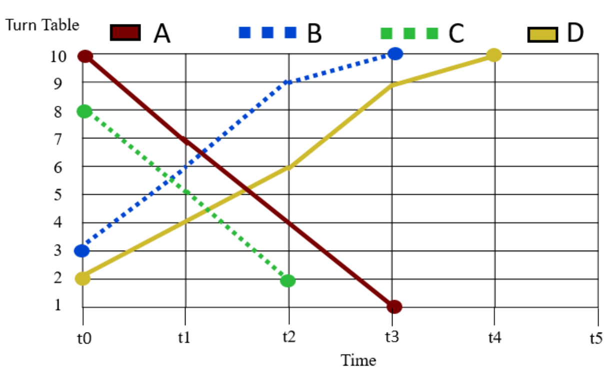

Fig. 13. The timeline for a complex pattern executed by four carriages on the Multi-Track Elevator System.

C. Travel Time and Total Distance Traveled

When observing a complex pattern, such as the ones demonstrated in the supplemental video and the one shown in Fig. 12, we can see that the individual carriages do not necessarily take the shortest possible path to reach their

TABLE I
CONTINUOUS MOTION SCHEDULE (CMS) VS. SHORTEST PATH TRAVELED (SPT)

Carriage Letter	A	B	C	D
Total Distance (mm) CMS	2540	2540	1956	1956
Total Distance (mm) SPT	1956	1956	1956	1956
Total Travel Time (seconds) CMS	27	27	27	27
Total Travel Time (seconds) SPT	36	36	27	27
Total Waiting Time (seconds) CMS	0	0	0	0
Total Waiting Time (seconds) SPT	27	0	9	9
Total Circuit Time (seconds) CMS	36	36	36	36
Total Circuit Time (seconds) SPT	54	27	36	36

desired location. A comparison can be made between the total time spent and distance traveled for situations in which a carriage is traveling the shortest distance pattern, Fig 14(b) as well as the continuous motion scheduling pattern, Fig 14(a).

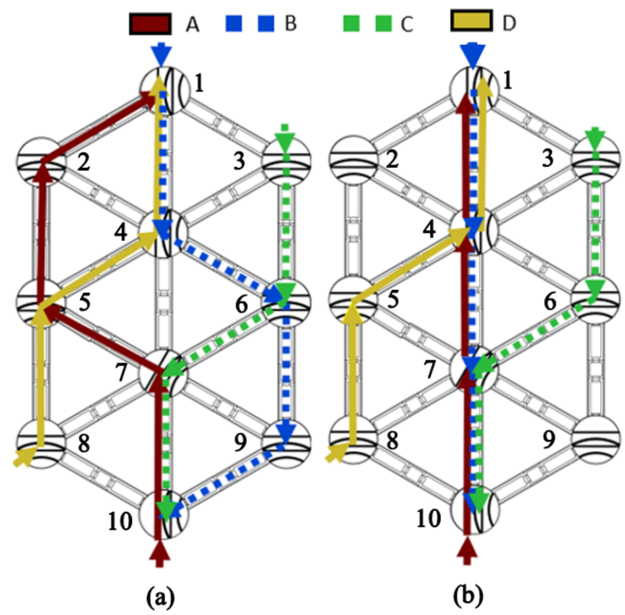

Fig. 14. A comparison of the (a) Continuous Motion Scheduling and (b) Shortest Path Traveled Scheduling of four carriages

Using the prototypes depicted in Figs. 8 and 9, the average travel time between the center of two turntables is approximately nine seconds. As shown in Table 1, the Continuous Motion Schedule (CMS) has a farther average distance traveled than the Shortest Path Traveled Schedule (SPT). However, the SPT takes 12 seconds longer to complete than the CMS. Therefore, if total travel distance is of lesser priority than total time, the Continuous Motion Schedule can be implemented. However, if the costs associated with continuous motion of the carriages are greater than those of time lost due to waiting, then the Multi-Track Elevator System carriages can be scheduled such that they take the shortest path.

The ability of carriages to take longer distance routes so that they may remain in continuous motion is critical when the optimal route of carriages with high priority tasks intersects with carriages assigned to lower priority tasks. The carriages assigned to the higher priority tasks are able to take the shortest possible path to their destination, while the carriages with less critical tasks are able to take a longer path, yet still make progress toward their destination [2].

VI. CONCLUSION

The MTE System has promise to be a significant advancement from current elevator standards available on the market today in terms of cargo transportation efficiency and its application to the e-commerce industry. Due to the ability to avoid routing conflicts, remain in continuous motion while avoiding collisions and the bi-directionality of each of the rails, the MTE system provides each carriage with a greater freedom of route choice and execution than that of the standard elevator and the CME system. Furthermore, the MTE system allows for carriages to reach their desired location at an overall faster normalized rate than the standard elevator and CME system. If distance traveled on the MTE system is of higher priority than the total time taken to complete the assigned tasks, the MTE system is able to schedule carriage paths such that they each take the shortest distance to their end goal at the expense of sacrificing total time to complete the task. However, if the total time executing tasks is to be minimized, then the MTE system is able to execute scheduled carriage paths that are the best compromise between total distance traveled and shortest path to destination. Using this method, each carriage remains in continuous motion and are constantly making progress towards their end goals, although they may not necessarily be taking the shortest possible route towards their goals. Thus, the MTES provides a highly adaptive scheduling capability.

Further investigations will be made into optimizing the real-time scheduling of multiple carriages as well as into localization techniques, such as cameras, location tags and indoor navigation systems, to track individual carriages.

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